Hurwitz–LL Compactification of Rerooting

Repository working draft

July 22, 2026

Abstract

We compactify the rerooting relation for normalized marked-root pencils by passing to marked admissible polynomial covers. The resulting quotient-stack description extends the generic degree-$N-2$ rerooting map, identifies its boundary pullbacks, and compares the normalized Stein factors and conductor square with the affine incidence constructions at arbitrary simultaneous root collisions. We also compute the degree $(N-2)N^{N-3}$ of the marked-zero- fiber Lyashko–Looijenga morphism and the caustic and Maxwell divisor classes on the root-boundary compactification.

1 Setup, provenance, and review status

Let H be a normalized degree-N seed with an exact double root at zero and otherwise simple zero-fiber points on the dense open under consideration. For

$$P_{H,s}(W) = H(W) - sW,$$

a critical point r has $s = H'(r)$ and critical value $H(r) - rH'(r)$. Thus the discriminant normalization is

$$r \longmapsto \big(H'(r), rH'(r) - H(r)\big).$$

The $N-2$ nonzero simple roots give the generic rerootings; collision strata are treated in the marked admissible-cover closure.

The admissible-cover technology cited below from Abramovich–Corti–Vistoli and Deopurkar supplies an appropriate proper compactification framework. It does not externally review the repository-specific restricted LL-degree, boundary-pullback, Picard, caustic, or Maxwell calculations. No external review of those calculations is recorded, and the statements below should be read with that provenance distinction explicit.

Proposition 1.1 (Hurwitz–LL compactification)**.** *For $P_{H,s}(W) = H(W) - sW$, the curve D_H is the pullback of the universal Lyashko–Looijenga point-in-branch-divisor incidence along $s \mapsto P_{H,s}$. The normalized seed locus has a proper compactification $\overline{\mathcal{S}}_N^{\mathrm{adm}}$ obtained as the normalization of its closure in the stack of marked admissible polynomial covers. The generic degree-$N-2$ rerooting relation extends to the proper marked/unmarked groupoid. On the collision-separating model it is a finite representable etale quotient-stack morphism. A total collision of m zero-fiber points has m-cycle monodromy on the special transverse slice $\epsilon = \delta^m$, while generic coarse divisorial inertia is a transposition.*

Proof. A critical point r of $P_{H,s}$ satisfies $s = H'(r)$, and its critical value is

$$P_{H,s}(r) = H(r) - rH'(r) = -t.$$

This proves the first assertion and recovers the normalization (6.1). On the exact-double distinct-root open,

$$H(W) = -\frac{W^2(W-1)\prod_{i=1}^{N-3}(W-\rho_i)}{\prod_{i=1}^{N-3}(1-\rho_i)},$$

so the labelled source is the configuration space of the $N-3$ points ρ_i on the thrice-marked rational curve. Its collision-separating compactification is $\overline{M}_{0,N}$; quotienting by S_{N-3} forgets their labels. Equivalently, the seed defines a polynomial Hurwitz cover with total ramification over infinity, a marked ramified point p_0 over zero, and a marked unramified point p_1 over zero. Taking the normalization of its closure in the proper admissible-cover stack gives $\overline{\mathcal{S}}_N^{\mathrm{adm}}$.[1]

Put $n = N-2$. Forgetting which of the n simple zero-fiber points is designated p_1, while retaining all of them as an unordered divisor, is the quotient morphism

$$[\overline{M}_{0,N}/S_{n-1}] \longrightarrow [\overline{M}_{0,N}/S_n]. \tag{6.15}$$

It is finite, representable, and etale of degree $[S_n : S_{n-1}] = n$, and restricts to the rerooting relation (6.8). On the cyclic slice $u^m = \epsilon$, normalization writes $\epsilon = \delta^m$ and cycles the m choices. For $m > 2$ this slice passes through the codimension-$(m-1)$ total-collision locus; the generic coarse discriminant instead has one pair collision and transposition inertia. Finally, the stable-map branch divisor extends in flat families, so the LL incidence extends over the closure.[2] Its finite part has degree $N-1$, after removing the fixed branch contribution $(N-1)[\infty]$. □

Proposition 1.2 (rerooting boundary pullback)**.** *Let π be (6.15). For any boundary divisor whose distinguished component contains exactly k of the n simple zero-fiber marks, write $\Delta_{k,\mathrm{in}}$ and $\Delta_{k,\mathrm{out}}$ according as p_1 belongs to that component or not. Then*

$$\pi^*[\Delta_k] = [\Delta_{k,\mathrm{in}}] + [\Delta_{k,\mathrm{out}}], \tag{6.16}$$

with restriction degrees k and $n-k$. After contraction to the coarse coefficient collision model, the generic discriminant pullback consists of one degree-one component of ramification index two and one unramified component of degree $n-2$.

Proof. The quotient-stack map (6.15) is etale, so both coefficients in (6.16) are one. Above an unordered boundary configuration, selecting one of the k points on the distinguished component gives the first restriction, while selecting one of the remaining $n-k$ gives the second. At a generic coefficient collision, choosing the double value has local square-root normalization and contributes $2 \cdot 1$ to the degree; choosing one of the $n-2$ other values is unramified. Thus the degree sum is $2 + (n-2) = n$. □

Remark 1.3. *The number $N-2$ here is the subgroup index $[S_n : S_{n-1}]$, not the degree of the classical LL morphism. The latter has degree N^{N-2} on the full space of monic centered degree-N polynomials.*[3] *The fixed pencil and the normalized seed locus have smaller dimensions, so an LL degree for their restriction requires a separately specified Hurwitz stratum and intersection problem.*

[1] For the normalization and twisted-cover interpretation of the admissible-cover stack, see D. Abramovich, A. Corti and A. Vistoli, *Twisted bundles and admissible covers*. Collision-allowing intermediate models are constructed in A. Deopurkar, *Compactifications of Hurwitz spaces*.

[2] B. Fantechi and R. Pandharipande, *Stable maps and branch divisors*.

[3] See S. Lando and D. Zvonkine, *Counting ramified coverings and intersection theory on spaces of rational functions I*, and the references there for the classical polynomial LL degree.

Theorem 1.4 (marked-zero-fiber LL degree)**.** *Let $\mathcal{A}_N^\circ$ be the exact-double normalized seed locus on which all critical values are simple. Put $n = N-2$, let $v_1, \dots, v_n$ be the nonzero critical values, and write*

$$\prod_{j=1}^{n}(V - v_j) = V^n + c_1 V^{n-1} + \cdots + c_n.$$

Then

$$\Lambda_N : \mathcal{A}_N^\circ \longrightarrow \mathbb{P}(1, 2, \dots, n), \qquad H \longmapsto [c_1 : \cdots : c_n]$$

is generically finite of degree

$$\boxed{\deg \Lambda_N = (N-2)N^{N-3}}.$$

If $\overline{\mathcal{A}}_N^{\mathrm{LL}}$ is the normalized graph closure in the marked admissible-cover compactification and $\eta = c_1(\mathcal{O}_{\mathbb{P}(1,\dots,n)}(1))$, then

$$\int_{\overline{\mathcal{A}}_N^{\mathrm{LL}}} \overline{\Lambda}_N^* \eta^{N-3} = \frac{N^{N-3}}{(N-3)!}.$$

Proof. The classical polynomial LL map has degree N^{N-2}. Generic precomposition by μ_N acts freely on its monic centered source and preserves every critical value. Quotienting by affine source isomorphism therefore leaves N^{N-3} covers. Requiring one specified critical value to be zero is a transverse target slice and does not change that count. Its fiber profile is $(2, 1^{N-2})$, so marking an unramified point over zero gives $N-2$ choices. The source conditions $p_0 = 0, p_1 = 1$ and the target condition $H'(1) = -1$ then choose a unique normalized representative. This proves the degree formula.

The weighted-projective stack satisfies $\int_{\mathbb{P}(1,\dots,n)} \eta^{n-1} = 1/n!$. Multiplying by the generic degree and using $n = N-2$ gives

$$\frac{(N-2)N^{N-3}}{(N-2)!} = \frac{N^{N-3}}{(N-3)!},$$

as asserted. □

Theorem 1.5 (caustic and Maxwell boundary classes)**.** *Put $n = N-2$ and work on $\overline{\mathfrak{B}}_N^{\mathrm{root}} = [\overline{M}_{0,n+2}/S_n]$. Its boundary-orbit classes satisfy the unique relation*

$$\sum_{k=1}^{n-1} k(n-k)\Delta_{0,k} = \sum_{k=2}^{n} k(k-1)\Delta_k^{\mathrm{free}}. \tag{6.17}$$

Thus

$$\mathcal{B}_n = \{\Delta_2^{\mathrm{free}}, \dots, \Delta_n^{\mathrm{free}}, \Delta_{0,1}, \dots, \Delta_{0,n-2}\}$$

is a basis of the rational Picard group. If $\mathcal{C}_N$ and $\mathcal{M}_N$ denote the closures of the caustic and Maxwell divisors, then in the redundant orbit presentation

$$\begin{aligned}
[\mathcal{C}_N] &= -\sum_{k=2}^{n}(k-1)(k-2)\Delta_k^{\mathrm{free}} + \sum_{k=1}^{n-1} k(n-k)\Delta_{0,k}, \\
[\mathcal{M}_N] &= -\frac{1}{2}\sum_{k=2}^{n}(k-1)(k-2)(k-3)\Delta_k^{\mathrm{free}} \\
&\quad + \frac{1}{2}\sum_{k=1}^{n-1} k(n-k)(n+k-2)\Delta_{0,k}.
\end{aligned} \tag{6.18}$$

In the basis $\mathcal{B}_n$ these become

$$[\mathcal{C}_N] = 2\sum_{k=2}^{n}(k-1)\Delta_k^{\mathrm{free}}, \tag{6.19}$$

and

$$\begin{aligned}[\mathcal{M}_N] = \frac{1}{2}\sum_{k=2}^{n}(k-1)\big((2n-3)k-(k-2)(k-3)\big)\Delta_k^{\mathrm{free}} \\ -\frac{1}{2}\sum_{k=1}^{n-2}k(n-k)(n-k-1)\Delta_{0,k}.\end{aligned} \tag{6.20}$$

Moreover $[\mathcal{C}_N] = 2\kappa_1$*.*

Proof. The logarithmic differential dH/H has residue vector $(2, 1^n, -N)$. Every proper partial sum is nonzero, so on a stable pointed source it has a nonzero residue at every node and its zero divisor avoids the nodes. If K is the relative dualizing class and D the marked divisor, the double-zero locus has class

$$\pi_*((K+D)(2K+D)) = 2\Big(\sum_i \psi_i - \delta\Big) = 2\kappa_1.$$

The symmetric boundary expansion of κ_1, together with the averaged Keel relation (6.17), gives (6.18)–(6.19) for the caustic.

For Maxwell, label the simple roots temporarily, normalize one of them to 1, and write $H = W^2\prod_i(W-a_i)$ and $K_H = H'/W$. On the interior,

$$\mathrm{Disc}_V\,\mathrm{Res}_W(K_H, V-H) = \mathrm{unit}\cdot\mathcal{C}_N^3\mathcal{M}_N^2.$$

A free collision of k roots gives orders $(k-1)(k-2)$, $(k-1)(k-2)(k-3)/2$, and $k(k-1)(k-2)$ for caustic, Maxwell, and the left side. If k roots tend to zero, the first two orders are $k(k-1)$ and $k(k-1)^2/2$. If $q = n-k$ roots tend to infinity, their pole orders are

$$q(q-1+2k), \qquad \frac{q}{2}\big((n-1)(q-1+2k)+k(k-1)\big).$$

Averaging over the n choices of normalized simple root gives the $\Delta_{0,k}$ coefficients in (6.18), while a free collision contributes the negative of its vanishing order. Finally (6.17) gives (6.20). The powers 3 and 2 have positive pullback orientation. On the stable target its discriminant pullback also contains target-boundary components; the pole terms above have merely been moved across the principal-divisor equality, so (6.18) does not suppress that boundary contribution. □

Theorem 1.6 (formal comparison at arbitrary root collisions)**.** *Pull the marked admissible-cover compactification of Proposition 1.1 back along the pencil $H - sW$ and mark a point over $-t$. The normalized Stein factor of this incidence is the canonical finite cover*

$$X_H = \mathrm{Spec}\,k[s,t,W]/(H(W)-sW+t),$$

and the normalized Stein factor of its critical-point incidence is $\widetilde{D}_H = \mathrm{Spec}\,k[r]$ with map (6.1). After contraction of all admissible source and target bubbles, their completed boundary charts and normalization–conductor square agree with the repository constructions, including at simultaneous collisions of arbitrarily many root clusters.

More explicitly, if the distinct multiple roots are ρ_i, of multiplicity m_i, and

$$e_i = m_i - 1, \qquad E = \sum_i e_i,$$

then the conductor in the completed normalization over $(s,t)=(0,0)$ *is*

$$\bar{\mathfrak{c}}_{(0,0)} = \bigoplus_i u_i^{e_i(E-1)} k[[u_i]]. \tag{6.21}$$

Proof. On the dense nondegenerate locus the marked-point incidence is $H(W)-sW+t=0$. This hypersurface is smooth, since its t-derivative is one, and is finite of degree N over $k[s,t]$. Its coordinate ring and the normalized Stein algebra of the proper admissible incidence are finite normal $k[s,t]$-algebras with the same generic function algebra.[4] By uniqueness of integral closure they coincide. The identical argument for the critical marking gives $k[r]$ and the map $r \mapsto (H'(r), rH'(r)-H(r))$.

At any primitive root α, put $W=\alpha+u$. Completion and elimination of t give

$$\widehat{\mathcal{O}}_{X_H,(0,0,\alpha)} = k[[s,u]], \qquad t = s(\alpha+u) - H(\alpha+u), \tag{6.22}$$

so Hensel separation makes simultaneous distinct clusters a product of the same regular charts on both sides. The reconstruction coordinates are the same rational functions in the common function field, so normality identifies their pole divisors and maximal regularity opens as well.

For a multiple root ρ_i, write $H(\rho_i+u)=c_iu^{m_i}+O(u^{m_i+1})$. On its discriminant branch, with $q_i = t-\rho_i s$, formula (6.1) gives

$$\operatorname{ord}_{u_i} s = e_i, \qquad \operatorname{ord}_{u_i} q_i = e_i + 1. \tag{6.23}$$

Thus its intrinsic conductor exponent is $e_i(e_i-1)$. Distinct roots give distinct tangent lines $t=\rho_i s$, hence $I(D_i,D_j)=e_ie_j$. Applying the adjunction conductor formula (6.5a) to $f=\prod_i f_i$ adds these pairwise intersections on branch i and yields

$$e_i(e_i-1) + \sum_{j\neq i} e_ie_j = e_i(E-1),$$

which proves (6.21). Since both constructions have the same finite normalization map, quotienting by this common conductor gives the same conductor square. The argument descends from geometric branches by Galois invariance. $\square$

[4] We use the Noetherian Stein factorization theorem in its relative-normalization form; see Stacks Project, Tag 03H0.